

DS701 Homework 2: Distances

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Name: _____ BU ID: _____

How this works. Answer **by hand on paper** (or a tablet with a stylus), show your work, then scan and upload to Gradescope, matching each problem to its pages. This is deliberate practice for the closed-notes quizzes, which are in the same style — so do it without a computer or an AI assistant. It is **participation-graded**: a serious attempt at every problem earns full credit, and worked solutions are released the day after the deadline. Budget 1–2 hours.

Covers: Lecture 02 (Distances and Similarity — metrics, ℓ_p distances, cosine, Hamming/Jaccard, feature scaling). Time series and DTW are *not* covered. Square roots may be left as $\sqrt{\cdot}$ with a one-decimal approximation next to them (e.g. $\sqrt{12} \approx 3.5$).

Problem 1: The metric decides who is close

Practices: Minkowski distances, cosine similarity, Hamming and Jaccard — and noticing that “closest” depends on the choice.

Three short documents are represented as word-count vectors over the vocabulary (*data*, *model*, *error*):

$$\mathbf{A} = (1, 1, 1), \quad \mathbf{B} = (3, 3, 3), \quad \mathbf{C} = (6, 1, 1).$$

- (a) For each of the three pairs (A,B), (A,C), (B,C) compute the Minkowski distance $\|\mathbf{x} - \mathbf{y}\|_p$ for $p = 1$ (Manhattan), $p = 2$ (Euclidean) and $p = \infty$ (Chebyshev) — nine numbers, arranged as a 3×3 table. Under each of the three distances, which pair is **closest**? Note where the answer changes.

pair	$p = 1$	$p = 2$	$p = \infty$
(A,B)	_____	_____	_____
(A,C)	_____	_____	_____
(B,C)	_____	_____	_____

Closest pair: $p = 1$: _____ $p = 2$: _____ $p = \infty$: _____

- (b) Compute the cosine similarity for the same three pairs (three decimals; $\sqrt{3} \approx 1.732$, $\sqrt{27} \approx 5.196$, $\sqrt{38} \approx 6.164$, $\sqrt{114} \approx 10.677$). Which pair is most similar under cosine? Explain in one sentence why cosine gives A and B the value it does even though every ℓ_p distance in (a) says they are some way apart, and say which of the two views is the right one if we want to know whether two documents are *about the same thing*.

$\cos(\mathbf{A}, \mathbf{B}) =$ _____ $\cos(\mathbf{A}, \mathbf{C}) =$ _____ $\cos(\mathbf{B}, \mathbf{C}) =$ _____

Most similar pair: _____

(c) Two pairs of shopping baskets, as sets of items:

- $P = \{\text{bread, milk, eggs, butter, cheese, apples, rice, coffee}\}$ and $Q = \{\text{bread, milk, eggs, butter, cheese, apples, rice, tea}\}$;
- $R = \{\text{bread, milk}\}$ and $S = \{\text{bread, jam}\}$.

For each pair compute the Hamming distance (the number of items in one basket but not the other, i.e. the size of the symmetric difference) and the Jaccard similarity $J_{\text{Sim}} = |x \cap y| / |x \cup y|$. Which pair does Hamming say is more alike, and which does Jaccard say? In one or two sentences, say which measure you would trust for “are these two shoppers similar?” and why.

pair	Hamming	J_{Sim}
(P,Q)	_____	_____
(R,S)	_____	_____

Problem 2: Is it a metric?

Practices: the metric axioms, especially the triangle inequality, and why they matter downstream.

Recall the four metric axioms: $d(x, x) = 0$; $d(x, y) > 0$ for $x \neq y$; $d(x, y) = d(y, x)$; and the triangle inequality $d(x, y) \leq d(x, z) + d(z, y)$ for all x, y, z .

- (a) Let $P = (0, 0)$, $Q = (3, 4)$, $R = (6, 0)$ in $\mathbb{R}^2$. Compute the three Euclidean distances $d(P, Q)$, $d(Q, R)$, $d(P, R)$ and verify the triangle inequality for this triple in all three ways (each distance is at most the sum of the other two). Then state, in one line, when the triangle inequality holds with **equality** for Euclidean distance, and give a triple of points that achieves it.

$$d(P, Q) = \underline{\hspace{2cm}} \quad d(Q, R) = \underline{\hspace{2cm}} \quad d(P, R) = \underline{\hspace{2cm}}$$

- (b) Cosine *distance* is $d_{\cos}(\mathbf{x}, \mathbf{y}) = 1 - \cos(\mathbf{x}, \mathbf{y})$. The lecture says it is **not** a metric. Prove it: give three explicit vectors $\mathbf{x}, \mathbf{z}, \mathbf{y}$ in $\mathbb{R}^2$ for which $d_{\cos}(\mathbf{x}, \mathbf{y}) > d_{\cos}(\mathbf{x}, \mathbf{z}) + d_{\cos}(\mathbf{z}, \mathbf{y})$, and show the numbers (use $\cos 45^\circ = 1/\sqrt{2} \approx 0.707$ if it helps). *Hint:* think about angles 0° , 45° , 90° .

$$\mathbf{x} = \underline{\hspace{2cm}} \quad \mathbf{z} = \underline{\hspace{2cm}} \quad \mathbf{y} = \underline{\hspace{2cm}}$$

- (c) Cosine distance fails a **second** axiom as well. Which one, and what is a two-vector example? (One line.)

- (d) In two or three sentences: why does it matter, for nearest-neighbor search or for clustering, whether the dissimilarity you use satisfies the triangle inequality? Concretely, what can you conclude about $d(q, b)$ from knowing $d(q, a)$ is small and $d(a, b)$ is large when d is a metric — and what goes wrong when it is not?

Problem 3: Same points, different units, different neighbor

Practices: why raw Euclidean distance is dominated by the largest-range feature, and what standardizing does.

A bank describes customers by two features: annual income (in dollars) and age (in years). A query customer q and four customers on file:

	income	age
q	50,000	30
P_1	52,000	60
P_2	60,000	32
P_3	45,000	28
P_4	51,000	45

To **standardize** a feature, replace each value v by its z -score $(v - \mu)/\sigma$, where μ and σ are the feature's mean and standard deviation. Use these values, taken from the bank's full customer base: income $\mu = 50,000$, $\sigma = 10,000$; age $\mu = 40$, $\sigma = 10$.

- (a) Using raw Euclidean distance, which of $P_1, \dots, P_4$ is the nearest neighbor of q ? (Squared distances are enough to rank; report them.) In one line: how much does the age difference contribute to any of these distances?

	P_1	P_2	P_3	P_4
d^2 from q	_____	_____	_____	_____

Nearest neighbor: _____

- (b) Standardize all five rows with the given μ, σ (write the z -scores), recompute the four squared Euclidean distances from q , and give the nearest neighbor now. Does it change?

	q	P_1	P_2	P_3	P_4
z_{income}	_____	_____	_____	_____	_____
z_{age}	_____	_____	_____	_____	_____
d^2 from q	—	_____	_____	_____	_____

Nearest neighbor: _____

- (c) Redo (a) with income measured in *thousands* of dollars (50, 52, 60, 45, 51) and age unchanged. Which neighbor wins? Then answer: which of the three answers should a data scientist trust as “the customer most like q ”, and why — what is wrong with the other two?

	P_1	P_2	P_3	P_4
d^2 from q	_____	_____	_____	_____

Nearest neighbor: _____ Trust: (a) / (b) / (c)

- (d) Standardizing is not always right. Give one situation where you would **not** standardize before computing distances, and one sentence on why (think about features that are already in the same meaningful units, or a feature whose spread *should* count).